COLOUR THEORY

Primary Colours

3 pigment colours that can not be mixed or formed by any combination of other colours. All other colours are derived from these 3 hues.

Secondary Colours

These are the colours formed by mixing two primary colours.

Tertiary Colours

These are the colours formed by mixing a primary and a secondary colour. That's why the hue is a two word name, such as blue-green, red-violet, and yellow-orange.

Analogous

Analogous colours sit next to one another on the colour wheel. These colours are in harmony with one another.

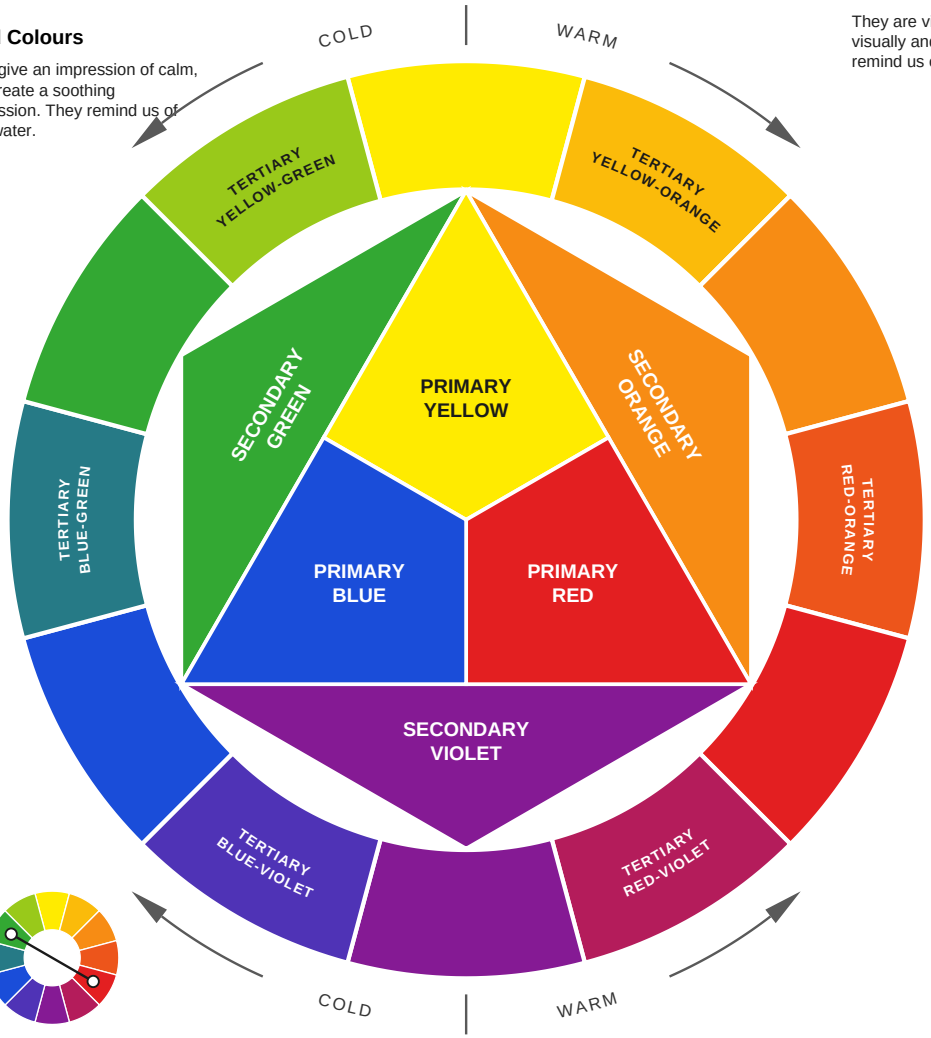
White, black and gray are considered to be neutral.

Cold Colours

They give an impression of calm, and create a soothing impression. They remind us of cold water.

Warm Colours

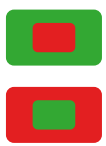
They are vivid and energetic, they are visually and emotionally exciting. They remind us of fire and heat.



Complementary Colours

Colours that are opposite each other on the colour wheel are considered to be complementary colours (example: red and green).

When put together, they appear more vivid than when apart.





Triadic colour scheme



Rectangle (tetradic) colour scheme



Split-Complementary colour scheme



Square colour scheme

Tint

A tint describes a colour that is mixed with white.

Tone

A tone describes a colour that is mixed with grey.

Shade

A shade describes a colour that is mixed with black.

Monochromatic

The term monochrome refers to the use of one colour or various shades and tints of one colour in a single form.

Meaning of Colour

(Western Culture)

RED:
Passion, Love, Fire, Anger, Blood

ORANGE:
Energy, Happiness, Vitality, Stimulation

YELLOW:
Sunshine, Happiness, Hope, Deceit

GREEN:
New Beginnings, Abundance, Nature

BLUE:
Sky, Calm, Responsible, Sadness, Sea

VIOLET:
Creativity, Royalty, Wealth, Ambition

BLACK:
Mystery, Elegance, Evil, Death, Power

GRAY:
Moody, Conservative, Formality

WHITE:
Purity, Cleanliness, Virtue, Innocence

BROWN:
Nature, Wholesomeness, Dependability

TAN OR BEIGE:
Conservative, Piety, Dull

CREAM OR IVORY:
Calm, Elegant, Purity

COLOR THEORY

COLOR MODELS

CMYK

Printing standard.
Mix of cyan, magenta, yellow and black inks.



RGB

Digital standard.
Mix of red, green and blue channels of light.



HSB

Variation of RGB.
Mix of hue, saturation and brightness.



GRAYSCALE

Intensity of black.
Uses only black intensity to create shade.

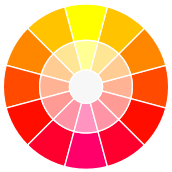




COLOR MEANINGS

- PASSION, LOVE, BLOOD, DANGER, STRENGTH
- ROMANTIC, FEMININE, LOVE, SENSITIVITY, EXCITING
- OPTIMISM, SUNSHINE, CONFIDENCE, ATTENTION, SUCCESS
- HEALTH, GROWTH, PEACE, VITALITY, WEALTH
- ENERGY, HAPPINESS, SUCCESS, WARMTH, CREATIVITY
- TRUST, SERENITY, PEACE, COMMUNICATION, SADNESS
- ROYALTY, MAJESTY, MYSTERY, SPIRITUALITY, LUXURY
- ORGANIC, HONEST, NATURAL, SIMPLE, WHOLESOME
- COMPROMISE, NEUTRAL, BALANCE, CONSERVATIVE, FRUSTRATION
- ELEGANCE, POWER, LUXURY, MYSTERY, SORROW
- PURE, SIMPLE, INNOCENCE, VIRGINITY, MINIMALISM

COLOR PROPERTIES



WARM



COOL



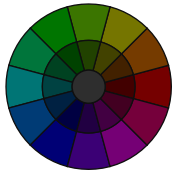
SATURATED



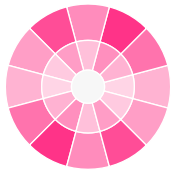
DESATURATED



BRIGHT



DARK



MONOCHROME

COLOR HARMONIES



ANALOGOUS



COMPLEMENTARY



TRIAD



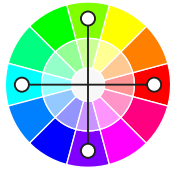
SPLIT COMPLEMENTARY



TETRAD

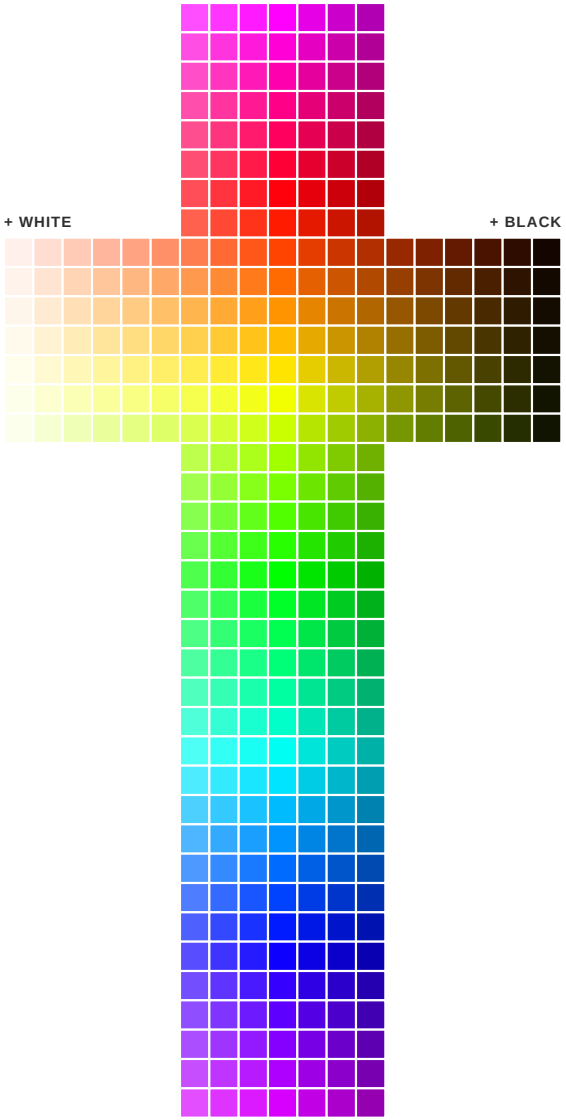


MONOCHROMATIC



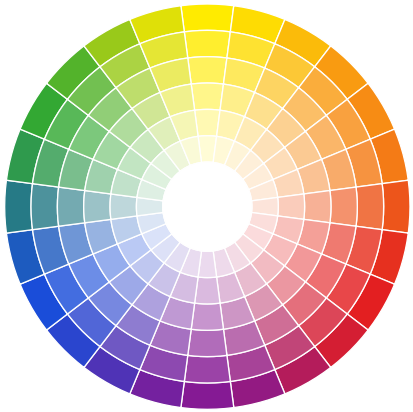
SQUARE

HUE & LIGHTNESS



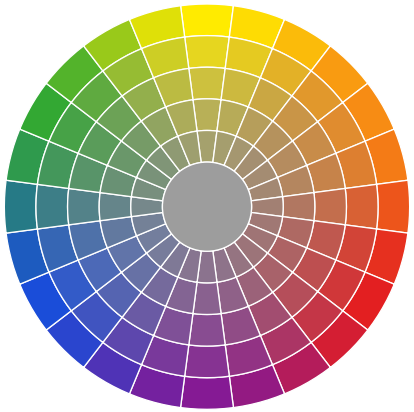
HUE × LIGHTNESS

Each row is a hue; each column adds white (left) or black (right). The long stem runs through every hue, the arm through the whole range from tint to shade.



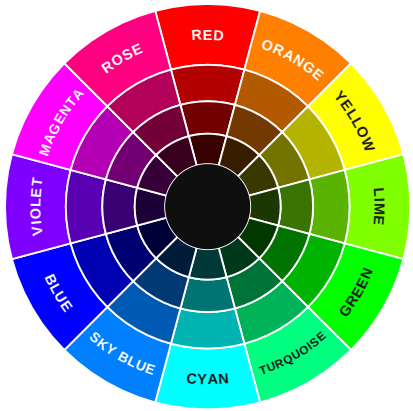
TINTS

24 hues of the RYB wheel, mixed with more and more white toward the centre.



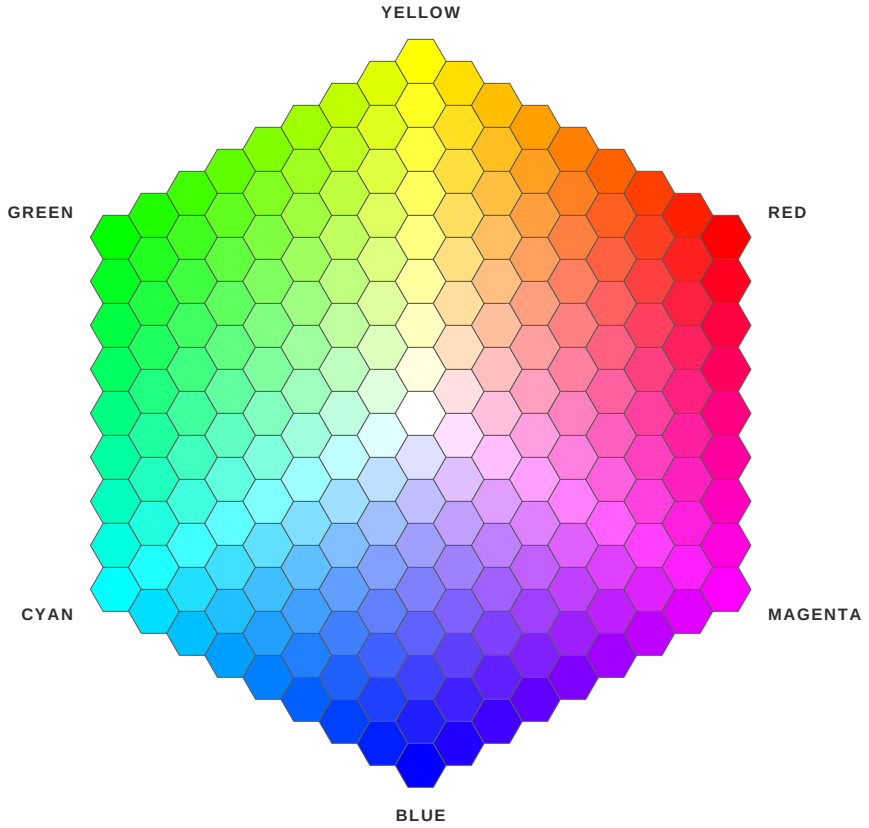
TONES

The same hues mixed with grey: softer, duller colours toward the centre.



SHADES

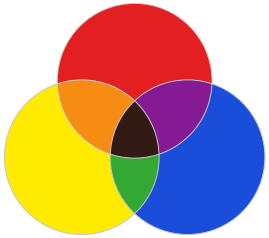
12 hues of light, darkened ring by ring by adding black.



THE RGB CUBE SEEN FROM ITS WHITE CORNER

Looking down the grey diagonal of the RGB cube, its three bright faces form a hexagon: white in the centre, the six pure hues at the corners, and every mixture of two channels in between. The hidden faces hold the darker colours.

COLOUR MIXING & CODES

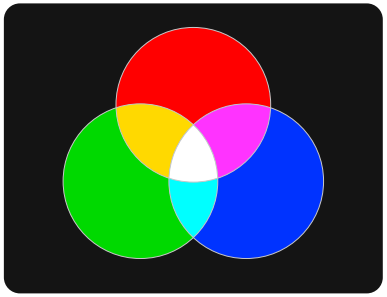


RYB

TRADITIONAL COLOUR

Mixing pigments: painting and fine art.
Subtractive: the three primaries make a dark brown.

Complementary: red – green

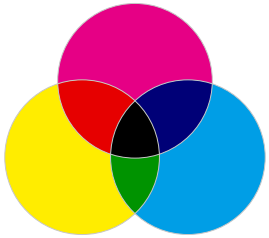


RGB

LIGHT

Mixing light: screens, cameras and photos.
Additive: the three primaries make white.

Complementary: red – cyan



CMYK

PRINTING

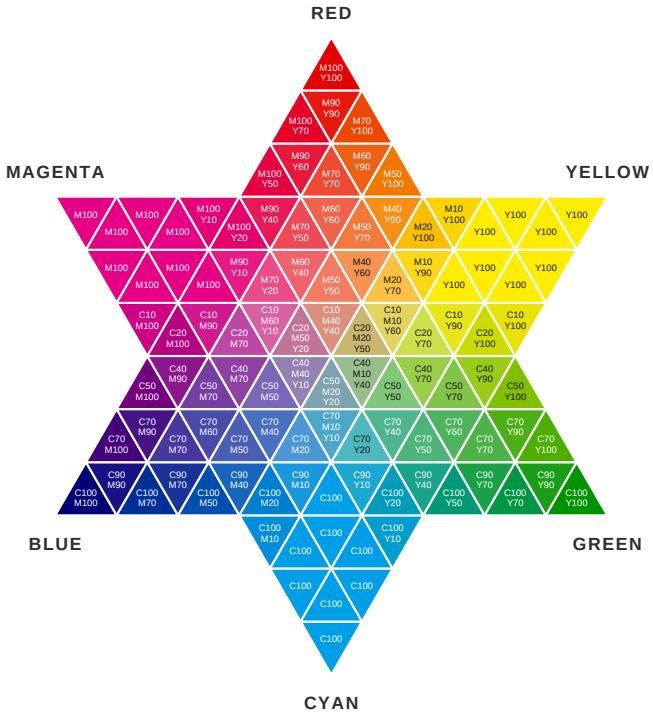
Mixing inks: commercial printing.
Subtractive: the three inks make a near-black, so black (K) ink is added. White is the paper.

Complementary: cyan – red

000	003	006	009	00C	00F	300	303	306	309	30C	30F	600	603	606	609	60C	60F
030	033	036	039	03C	03F	330	333	336	339	33C	33F	630	633	636	639	63C	63F
060	063	066	069	06C	06F	360	363	366	369	36C	36F	660	663	666	669	66C	66F
090	093	096	099	09C	09F	390	393	396	399	39C	39F	690	693	696	699	69C	69F
0C0	0C3	0C6	0C9	0CC	0CF	3C0	3C3	3C6	3C9	3CC	3CF	6C0	6C3	6C6	6C9	6CC	6CF
0F0	0F3	0F6	0F9	0FC	0FF	3F0	3F3	3F6	3F9	3FC	3FF	6F0	6F3	6F6	6F9	6FC	6FF
900	903	906	909	90C	90F	C00	C03	C06	C09	C0C	C0F	F00	F03	F06	F09	F0C	F0F
930	933	936	939	93C	93F	C30	C33	C36	C39	C3C	C3F	F30	F33	F36	F39	F3C	F3F
960	963	966	969	96C	96F	C60	C63	C66	C69	C6C	C6F	F60	F63	F66	F69	F6C	F6F
990	993	996	999	99C	99F	C90	C93	C96	C99	C9C	C9F	F90	F93	F96	F99	F9C	F9F
9C0	9C3	9C6	9C9	9CC	9CF	CC0	CC3	CC6	CC9	CCC	CCF	FC0	FC3	FC6	FC9	FCC	FCF
9F0	9F3	9F6	9F9	9FC	9FF	CF0	CF3	CF6	CF9	CFC	CFF	FF0	FF3	FF6	FF9	FFC	FFF

WEB-SAFE PALETTE

216 colours whose red, green and blue channels each take one of six values: 00, 33, 66, 99, CC or FF. The three-digit code #F93 is short for #FF9933. Each block has one red value; rows change green and columns change blue.



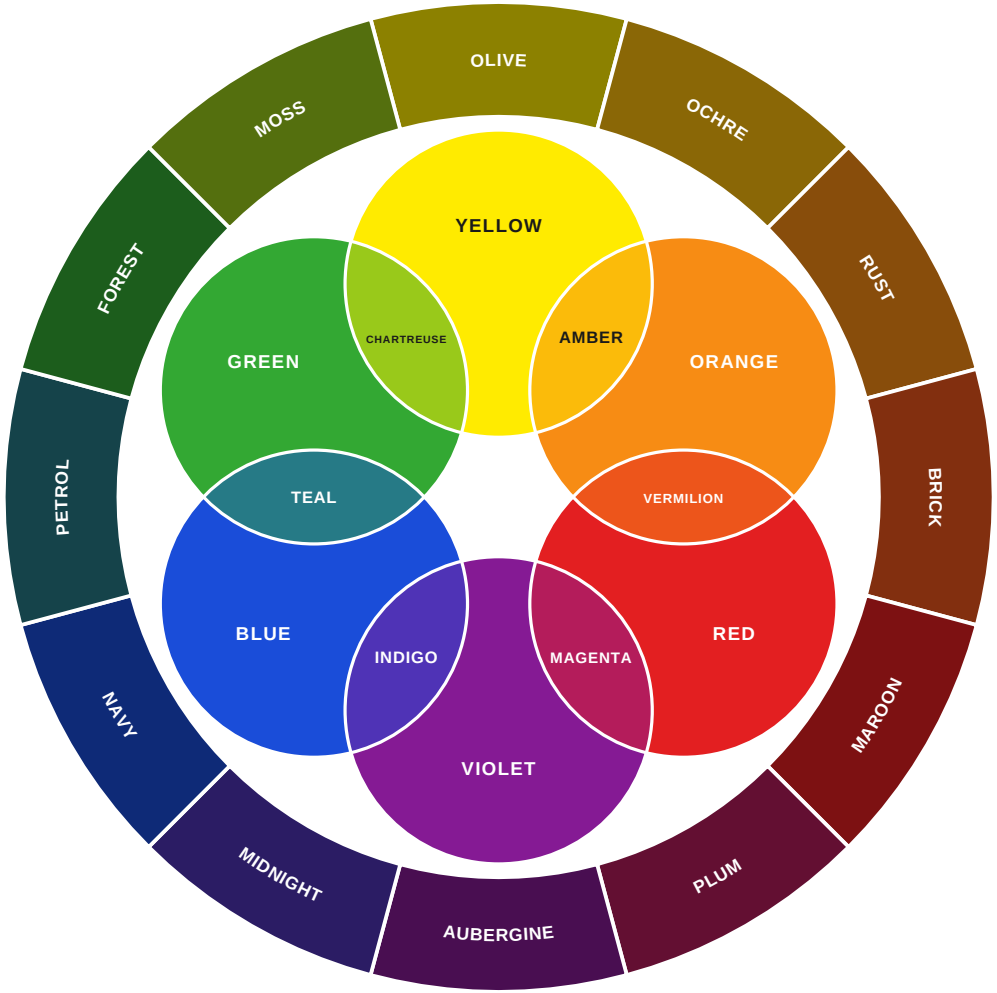
CMYK INK PERCENTAGES

Each triangle gives the percentages of cyan (C), magenta (M) and yellow (Y) ink that print it. Two inks at full strength make red, green or blue.

NAMED COLOURS

THE COLOUR WHEEL

Six overlapping circles of the RYB wheel: where two meet, their mixture appears. The outer ring shows each hue darkened with black.



CSS NAMED COLOURS

Colour names of the web (CSS keywords) with their hex codes, grouped by colour family and ordered by hue.

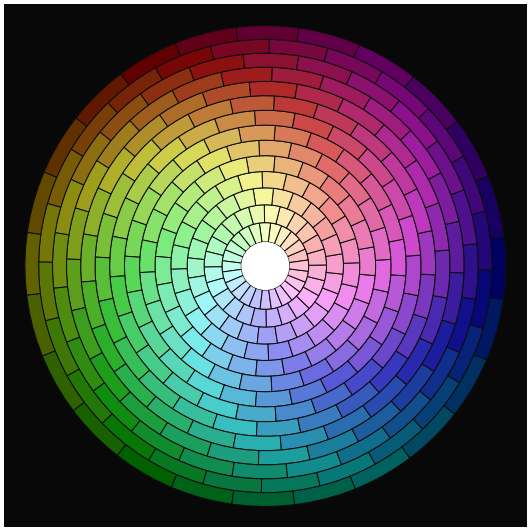


COLOUR & EMOTION

PLUTCHIK'S WHEEL OF EMOTIONS

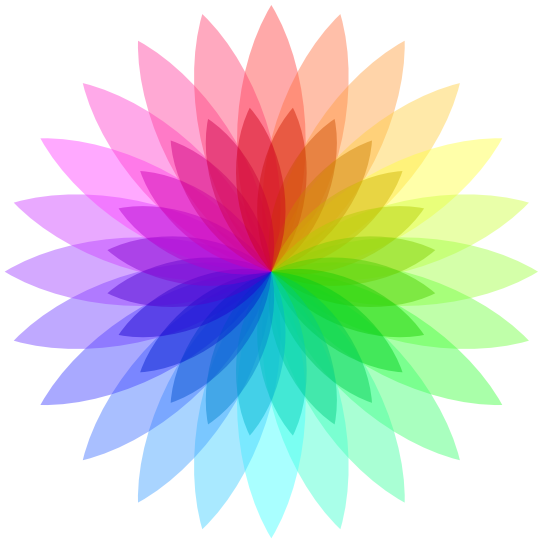
Eight basic emotions: stronger toward the centre, milder toward the tips. Between the petals, the feelings that combine two neighbours.





SPIRAL OF TINTS AND SHADES

Each ring is turned a little further than the one inside it, so hue and lightness sweep round in a spiral, from a white core to dark edges.

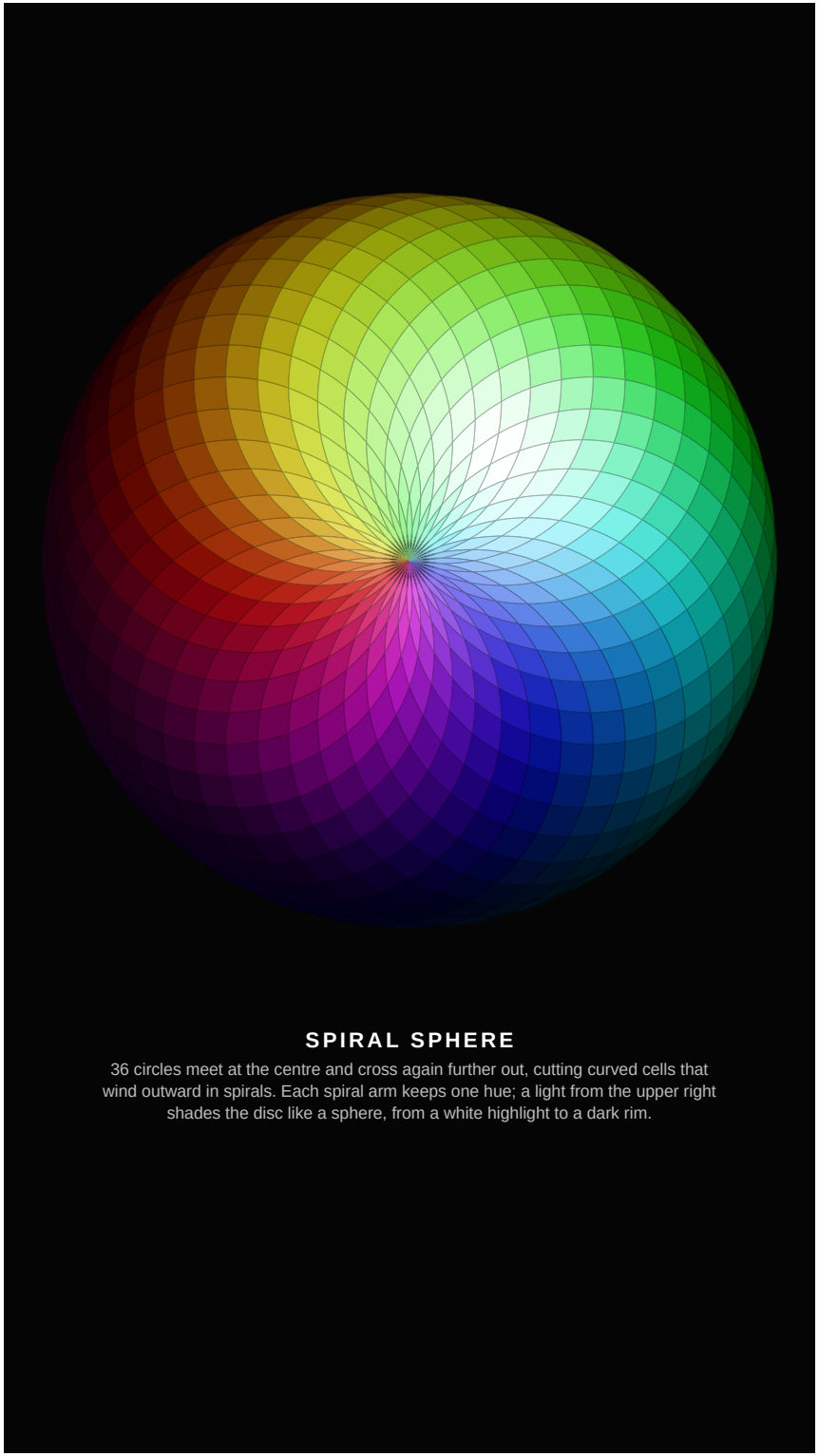


TRANSLUCENT PETALS

Two crowns of 24 half-transparent petals: where petals overlap, their colours combine into deeper ones.

THE COLOR WHEEL

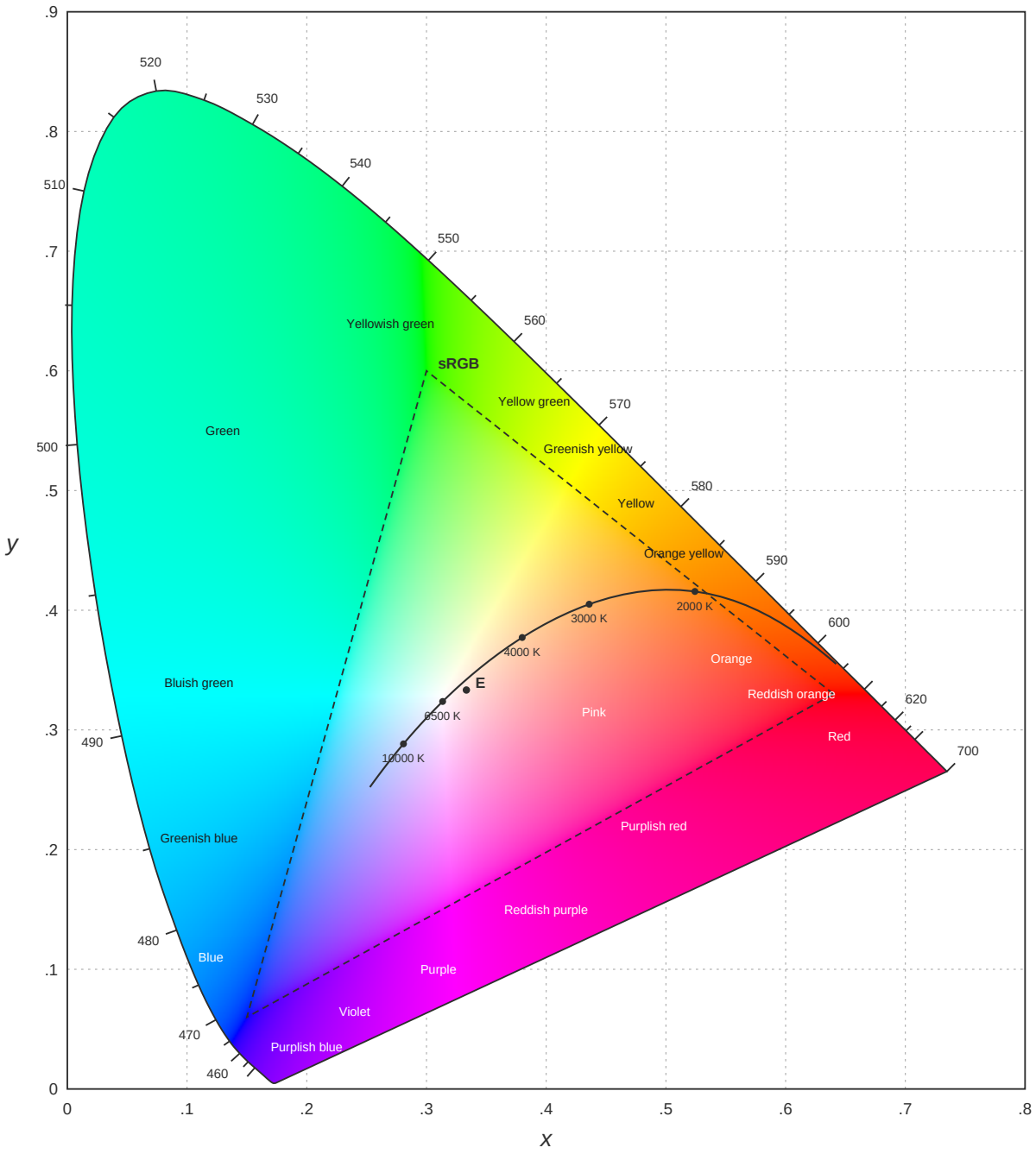




SPIRAL SPHERE

36 circles meet at the centre and cross again further out, cutting curved cells that wind outward in spirals. Each spiral arm keeps one hue; a light from the upper right shades the disc like a sphere, from a white highlight to a dark rim.

THE SPACE OF VISIBLE COLOURS



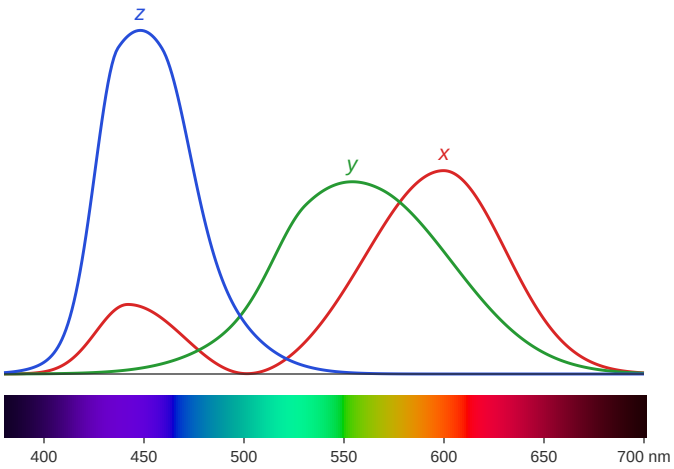
CIE 1931 CHROMATICITY DIAGRAM

Every colour the eye can see, at equal brightness, placed by its chromaticity coordinates x and y. Pure colours of a single wavelength lie on the curved edge; the straight base joins red and violet and holds the purples, which no single wavelength can produce. Inside are mixtures, paler toward white. A screen shows only the colours inside its triangle, so the fill here is an approximation.

- Spectral locus: one wavelength (nm)
- Line of purples: red + violet
- Planckian locus: colour of a glowing body (K)
- sRGB: the colours a screen can show
- E: equal-energy white

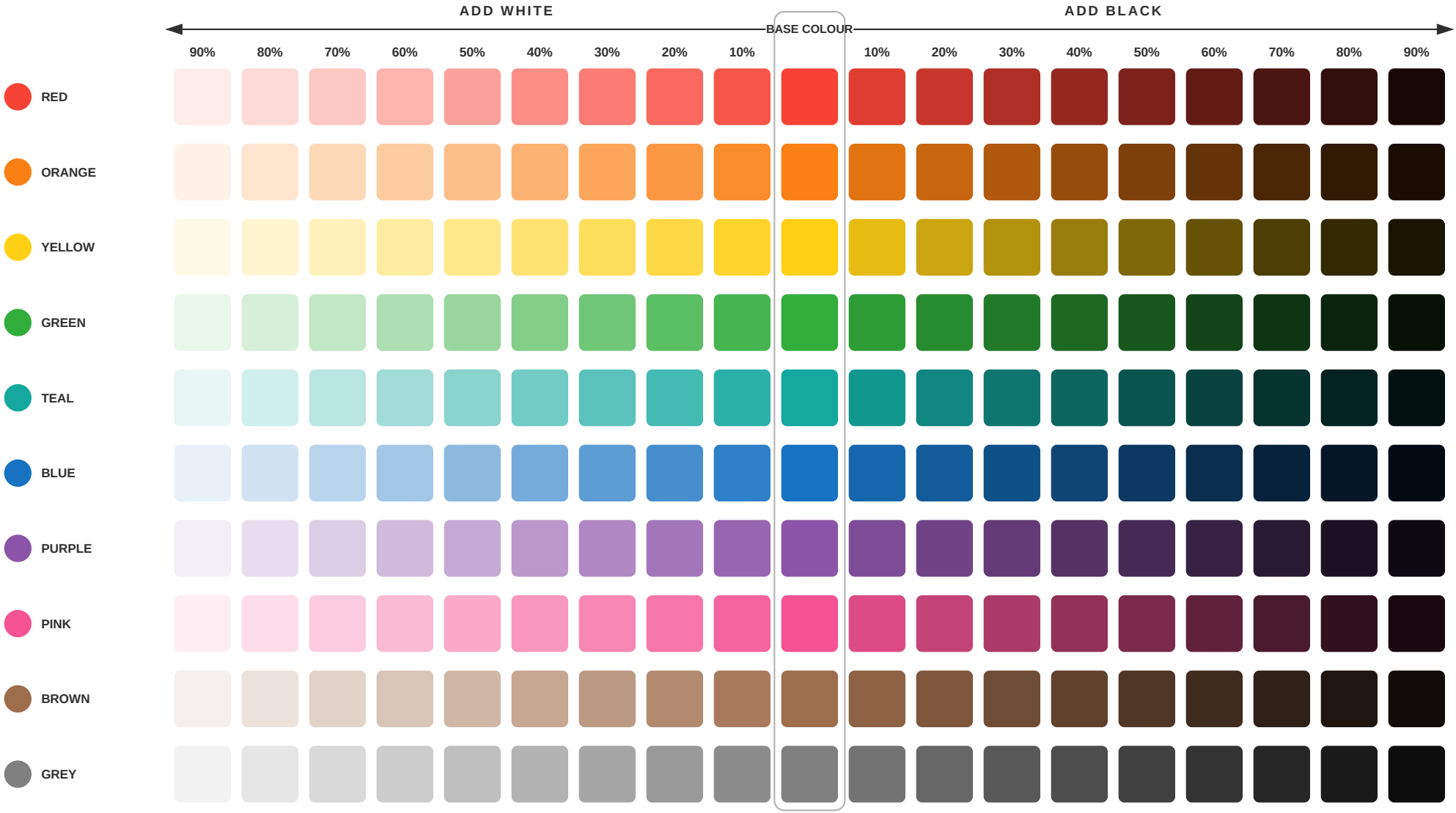
THE VISIBLE SPECTRUM

Bottom: wavelengths from 380 to 700 nm as screen colours. Top: the three colour-matching functions x, y, z, which turn any light into the X, Y, Z values behind the diagram.



TINTS AND SHADES

A tint adds white to a base colour; a shade adds black.

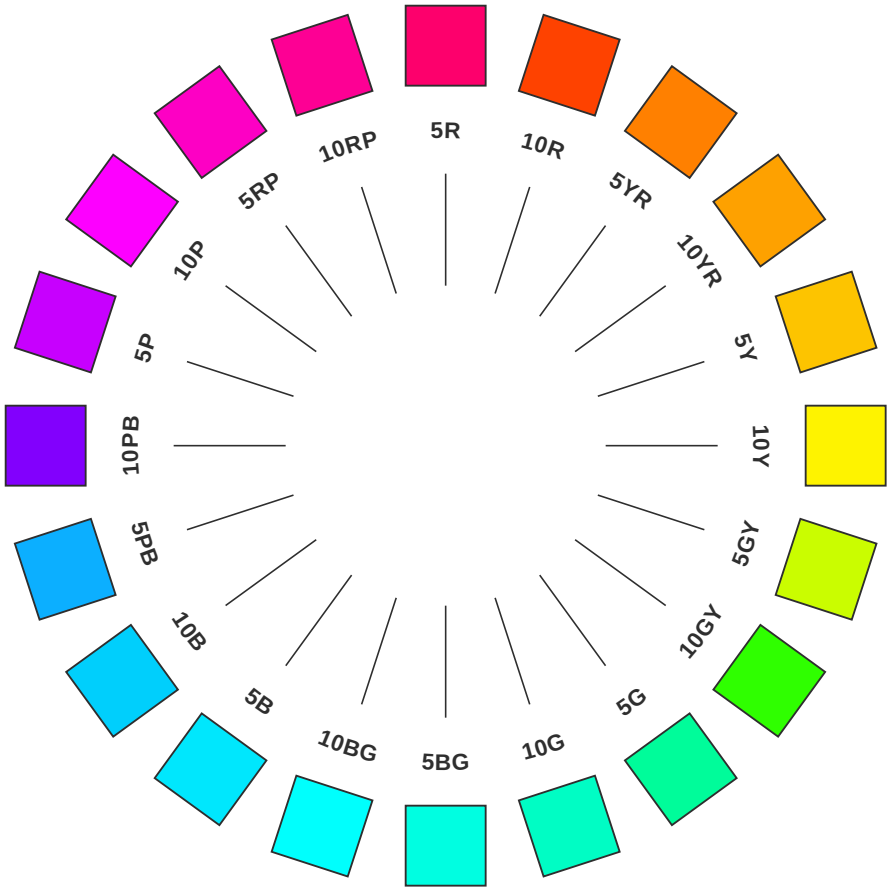


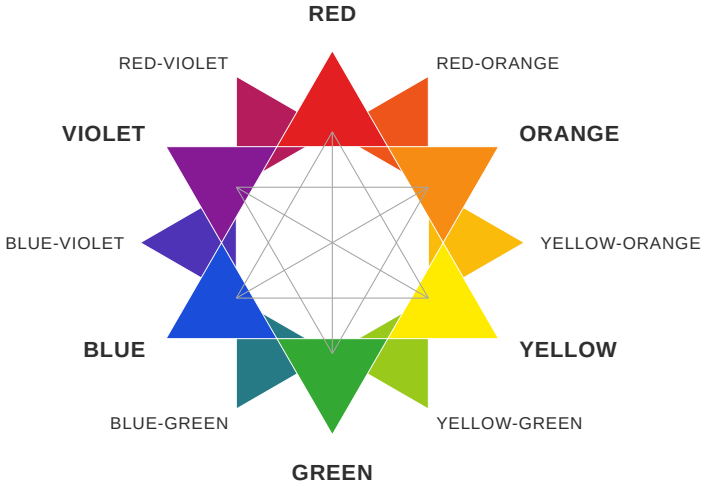
Tints are lighter and softer versions of a colour, shades are darker ones. Together they add depth, contrast and richness to any design.

HUE CIRCLES

MUNSELL HUE CIRCLE

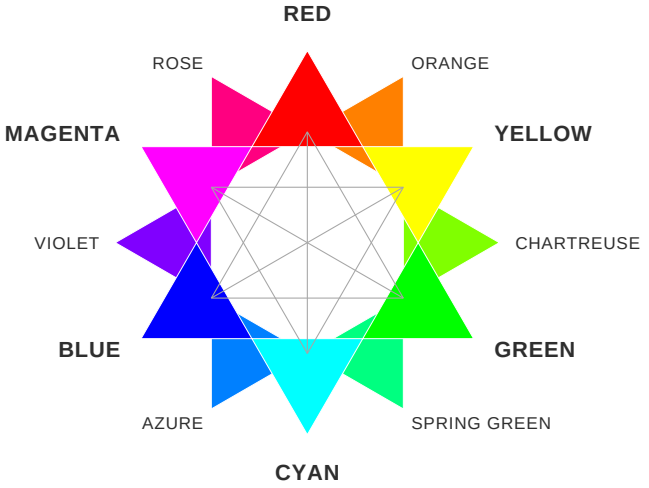
Five principal hues (R, Y, G, B, P) and five in between (YR, GY, BG, PB, RP), each split into steps 1 to 10; shown here at 5 and 10. Each swatch is the most saturated screen colour of that hue.





RYB

The painter's wheel: primaries red, yellow and blue on one triangle, secondaries on the other, tertiaries in the notches between them.



RGB

The wheel of light: primaries red, green and blue; their mixtures yellow, cyan and magenta; and six more colours in between.